

DUNGEY'S AVERAGED A_α RITT-PROBABILITY CONJECTURE

FA.BANACH.001 LITERATURE-STATUS PACKET

STATUS

This is a literature-already-answered packet. The source paper is Nick Dungey's *Subordinated discrete semigroups of operators*, arXiv:0801.4557. Dungey defines, for $\varepsilon \in (0, 1]$,

$$B = \varepsilon^{-1} \int_0^\varepsilon A_\alpha d\alpha$$

and conjectures on PDF page 19 that B belongs to the class $\mathcal{A}$ of Ritt probabilities.

Theorem 6.1 Fix $\varepsilon \in (0, 1]$ and set

$$B := \varepsilon^{-1} \int_0^\varepsilon d\alpha A_\alpha \in \mathbb{P}(\mathbb{Z}^+) \quad (23)$$

(with A_α defined by (19)). For each $\beta \in (0, 1)$ put $B_\beta := \sum_{k \geq 0} A_\beta(k) B^{(k)}$. Then $B_\beta \in \mathcal{A}$, and

$$\lim_{\xi \rightarrow 0} \frac{|\xi|^\alpha}{|1 - \widehat{B}_\beta(\xi)|} = 0 \quad (24)$$

for each $\alpha \in (0, 1)$.

Integration of the estimate (13) yields a bound $|1 - \widehat{F}(\xi)| \leq c'|\xi|^\alpha$ for all $\xi \in [-\pi, \pi]$; thus (24) shows that the probability $F = B_\beta$ cannot satisfy (13) for $\alpha \in (0, 1)$.

We point out that (23) has a formal similarity to a construction of [19] (see also [16] Section 2), where a Ritt operator is produced by integrating certain fractional Volterra operators.

Proof of Theorem 6.1. Clearly (23) defines an element $B \in \mathbb{P}(\mathbb{Z}^+)$. Since $A_\beta \in \mathcal{A}$, Proposition 4.2 implies that $B_\beta \in \mathcal{A}$.

Using (20) one computes

$$1 - \phi_B(w) = \varepsilon^{-1} \int_0^\varepsilon d\alpha (1 - w)^\alpha = \varepsilon^{-1} [\text{Log}(1 - w)]^{-1} ((1 - w)^\varepsilon - 1) \quad (25)$$

for $w \in \overline{\mathbb{D}}$; here $\text{Log}: \mathbb{C} \setminus (-\infty, 0] \rightarrow \mathbb{C}$ is the principal branch of the logarithm, with the additional convention $(\text{Log}(0))^{-1} = 0$. Replacing w by $e^{-i\xi}$ in (25), because of the logarithmic factor one sees that

$$\lim_{\xi \rightarrow 0} \frac{|\xi|^\alpha}{|1 - \widehat{B}(\xi)|} = 0$$

for each $\alpha \in (0, 1)$. But the definition of B_β implies that

$$\phi_{B_\beta}(w) = \sum_{k \geq 0} A_\beta(k) (\phi_B(w))^k = \phi_{A_\beta}(\phi_B(w)) = 1 - (1 - \phi_B(w))^\beta,$$

hence $1 - \widehat{B}_\beta(\xi) = (1 - \widehat{B}(\xi))^\beta$, and (24) follows. $\square$

We conjecture that B defined by (23) is an element of $\mathcal{A}$, for any $\varepsilon \in (0, 1]$. Evidence for this conjecture is that the generating function ϕ_B satisfies the sectorial condition in [9], as is easily verified from (25).

ANSWER SOURCE

Alexander Gomilko and Yuri Tomilov, *On discrete subordination of power bounded and Ritt operators*, arXiv:1504.00563, explicitly identify this as a question left open by Dungey. In their notation,

$$h_\varepsilon(\lambda) = 1 - \frac{1}{\varepsilon} \int_0^\varepsilon (1 - \lambda)^\alpha d\alpha = 1 - \frac{(1 - \lambda)^\varepsilon - 1}{\varepsilon \log(1 - \lambda)}.$$

Theorem 7.1 proves a geometric criterion for regular Hausdorff functions: if $1 - h(\mathbb{D})$ lies in a sector $\overline{\Sigma}_\gamma$, then $h(T)$ is Ritt for every power-bounded operator T . Example 7.4 verifies this criterion for h_ε , and concludes that $h_\varepsilon(T)$ is Ritt of angle $\varepsilon\pi/2$ for $0 < \varepsilon < 1$. The paragraph immediately following also covers the endpoint h_1 .

bounded operator T on X and any $\alpha \in (0, 1)$ the operator $L_{1+\alpha}(T)$ is Ritt of angle $\beta(\alpha)$.

However, we can be more precise here. By combining Theorem 7.1 with Proposition 10.1 from Appendix B, we conclude that $L_{1+\alpha}(T)$ is of angle $\beta(\alpha) = \alpha\pi/2$.

Example 7.4. Let for a fixed $\epsilon \in (0, 1)$,

$$(7.14) \quad h_\epsilon(\lambda) := 1 - \frac{1}{\epsilon} \int_0^\epsilon (1-\lambda)^\alpha d\alpha = 1 - \frac{(1-\lambda)^\epsilon - 1}{\epsilon \log(1-\lambda)}, \quad \lambda \in \mathbb{D}.$$

The function h_ϵ extends holomorphically to $\mathbb{C} \setminus (-\infty, 0]$, and denoting the extension by the same symbol we infer that

$$f_\epsilon(\lambda) := 1 - h_\epsilon(1-\lambda) = \frac{\lambda^\epsilon - 1}{\log \lambda^\epsilon}, \quad \lambda > 0,$$

belongs to $\mathcal{CBF}$ as a composition of the complete Bernstein functions $(\lambda - 1)/\log \lambda$ and λ^ϵ . If $\lambda \in \mathbb{C}_+$, then $\lambda^\epsilon \in \overline{\Sigma}_{\epsilon\pi/2}$, hence in view of the integral representation in (7.14), $f_\epsilon(\mathbb{C}_+) \in \overline{\Sigma}_{\epsilon\pi/2}$. Thus, by Example 3.9 b), the function h_ϵ is regular Hausdorff and

$$(1 - h_\epsilon)(\mathbb{D}) \subset \overline{\Sigma}_{\epsilon\pi/2}.$$

By Theorem 7.1 we conclude that if T is a power-bounded operator on X , then $h_\epsilon(T)$ is Ritt of angle $\epsilon\pi/2$. This settles a conjecture posed in [23, p. 1735]. Note that Theorem 5.1 in [23] can also be treated in a similar way, but we leave the details to the interested reader.

IDENTIFICATION

Duney's probability B has generating function

$$\phi_B(\lambda) = h_\epsilon(\lambda).$$

Duney's Theorem 3.1 characterizes $B \in \mathcal{A}$ by the equivalent operator statement that $\Psi(B; T) = h_\epsilon(T)$ is a Ritt operator for every power-bounded T . Gomilko–Tomilov prove exactly this statement and explicitly state that it settles Duney's conjecture.

REFERENCES

REFERENCES

- [1] N. Duney, *Subordinated discrete semigroups of operators*, arXiv:0801.4557.
- [2] A. Gomilko and Y. Tomilov, *On discrete subordination of power bounded and Ritt operators*, arXiv:1504.00563.